

# A grounded NLP system for advisor job-screening decisions

Real postings -> retrieved evidence -> role-fit recommendation -> human review

## Final deliverable

Runs end-to-end  
Uses real data  
Reports metrics  
Flags risk

# The advisor decision is the business problem

Career advisors need to decide which postings to forward, hold, or investigate.

100k

real project postings

80k

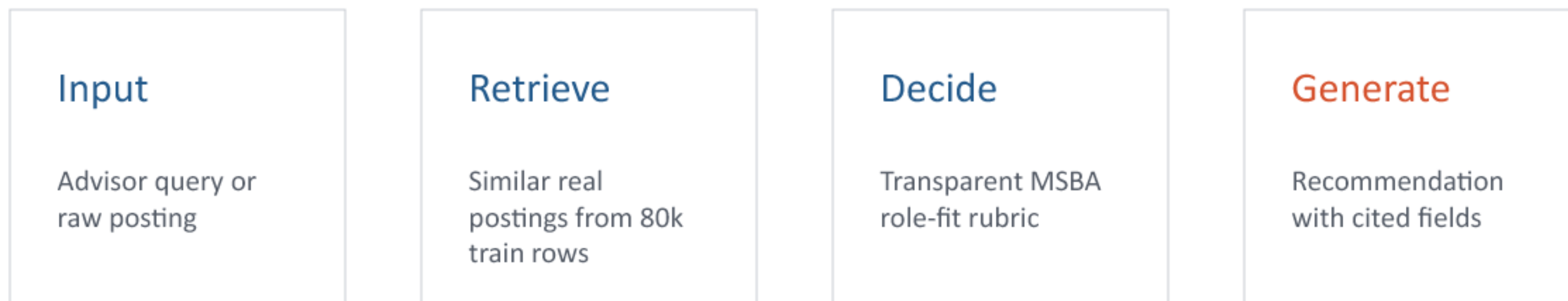
retrieval corpus

4

triage labels

The system is useful only if it gives evidence, not just a label.

# The system uses retrieval to keep recommendations grounded



Generation is constrained: it can cite retrieved fields and must flag missing authorization evidence.

# The live run returns a recommendation and evidence

**Input query**

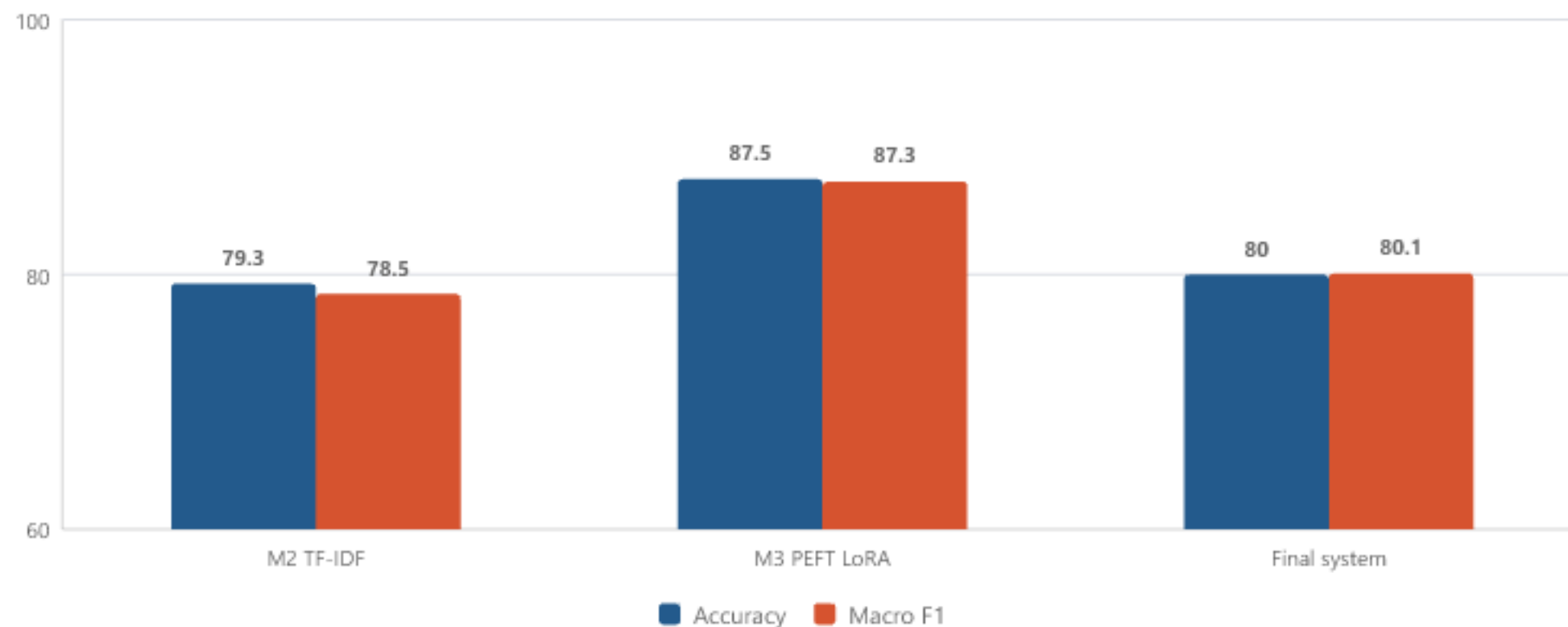
Find entry-level US analytics or business analyst opportunities for MSBA international students with SQL, Python, Excel, Tableau, or Power BI skills.

**Output**

Review retrieved postings first; they match analytics role-fit signals.

Evidence includes title, company, location, skills, and label. Authorization evidence remains a required human-review item.

# The final system slightly improves the baseline and adds evidence



## Final metric

80.0% accuracy

80.1% macro F1

The system adds grounded evidence and a refusal to infer work authorization.

# The edge case is authorization, not analytics fit

## Edge input

A student asks whether an employer definitely sponsors OPT or H-1B, but the posting has no authorization text.

## System response

Do not infer eligibility. Show retrieved evidence and route authorization to advisor review.

This is the key governance guardrail for international-student use.



# The business case is faster screening with safer escalation

## Time

Reduce first-pass review by surfacing likely matches and evidence.

## Quality

Separate role-fit recommendation from authorization uncertainty.

## Governance

Keep the advisor in the loop for missing evidence and edge cases.

# The next step is better labels and real authorization text

## Recommendation

Pilot with de-identified Career Center postings and collect advisor corrections.

## What to measure next

High-fit precision, authorization extraction accuracy, and human-review workload reduction.

The system is ready as a decision-support prototype, not an automated gatekeeper.